

Research Article

Tuning the linear and nonlinear optical absorption properties of ZnS/hydrochar nanocomposites by concentration of nanoparticles

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ABSTRACT

In this article, the effects of nanoparticles aggregates and concentration on linear and nonlinear optical absorption properties of ZnS and ZnS/carbon-based materials in polymethyl methacrylate (PMMA) matrix have been investigated. Carbon-based materials were synthesized via the hydrothermal carbonization method. Hazelnut shell was preferred as a feedstock of carbon material due to its high production capacity in Turkey. Thin films consisting of 10, 20, and 30 wt % nanoparticles were obtained to discover the effect of weight concentration on optical properties. The linear and nonlinear absorption properties were characterized by UV–Vis spectroscopy technique and open aperture Z-scan measurements with 4 ns pulses at 532 nm. A theoretical model containing one photon, two photon and free carrier absorption and their saturations were used for the fitting of the experimental data. The band gap decreased with the increasing of the weight concentration of nanoparticles in PMMA. Z-scan experiment results showed that the highest effective nonlinear absorption coefficient was obtained for the thin films with 30 wt% nanoparticle concentration which had also the smallest bandgap energy. The effect of nanoparticles aggregates on the band gap energy and nonlinear absorption was also observed. The transmission electron microscopy (TEM) results indicated that the addition of carbon caused the aggregation.

1. Introduction

Zinc sulphide (ZnS) is one of the most attractive materials among II-VI group compound semiconductor. It has a cubic and a hexagonal form. ZnS has been used for a range of applications due to its direct and wide band gap energy, high refractive index, and so on. It has potential to be used for light emitting diode, electroluminescence devices, photovoltaic cells, IR detectors and solar energy [1–6]. Many properties of ZnS, such as physical, chemical and structural, can be enhanced by decreasing the particle size to the nanoscale [7]. Different synthesis methods are available to prepare nanoparticles such as solvothermal method, hydrothermal method, gamma-irradiation and microwave irradiation [8]. The band gap can be adjusted and nonlinear optical properties can be changed depending on the nanoparticle concentration of the nanocomposite material obtained by mixing polymer and nanoparticle.

Hydrothermal carbonization (HTC) is one of the thermochemical conversion techniques which allows preparation of carbon materials (hydrochar) with semi-graphitic carbon skeleton and surface oxygen

functionalities. This technique is quite useful to obtain carbon materials from the different kinds of biological materials [9,10]. In the present study, HTC is used for the synthesis of hazelnut shell based carbonaceous nanoparticles (HBC). Carbonaceous nanoparticles (CNP) have a wide field of application such as water purification, electrode materials, catalysts and solid fuel [11–13]. The linear and nonlinear optical absorption behaviors of PMMA/ZnS and PMMA/ZnS-HBC nanocomposite films were reported. To investigate concentration effect, ZnS and HBC/ZnS nanoparticles with various concentrations (10, 20 and 30 wt %) were added to PMMA/cyclohexanone. Open aperture Z-scan technique was used to understand the nonlinear absorption properties of all samples.

2. Experimental section

Synthesis of zinc sulphide nanoparticles performed in Teflon-liner (50 mL) using CEM MARS-5 (frequency 2.45 GHz, maximum power 1200 W) microwave oven. 0.1 M zinc acetate dihydrate ($\text{Zn}(\text{CH}_3\text{COO})_2 \cdot 2\text{H}_2\text{O}$ Merck, %99) and 0.12 M thioacetamide were dissolved in 50 mL

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purified water. Transparent solution was observed after stirring 10 min and then placed into microwave oven. The reaction temperature was set 100 °C (600W) for 10 min. After the period synthesized nanoparticles were centrifuged, washed and dried under vacuum at 105 °C overnight.

The hydrothermal carbonization (HTC) at 260 °C for 4 h was used to prepare the hazelnut shell based carbonaceous nanoparticles. Details of the HTC procedure were given in our previous study [11]. In order to obtain ZnS-HBC composite material, ZnS and HBC were mixed with 3:1 wt ratio in ethanol/water solution at 1500 rpm for 24 h. After the period, solid product was collected via vacuum filtration and rinsed with water. The obtained product were dried in an oven at 90 °C overnight [14].

For the fabrication of ZnS and HBC/ZnS nanocomposites, 1 g PMMA was dissolved in 10 ml cyclohexanone by placing in a sonic bath for 48 h. After that, nanoparticles were added to this solution and the magnetic stirrer was used to obtain homogeneous distribution. Finally, nanocomposite films were prepared by SCS-Spin Coat G3P on fused silica substrate and were kept in vacuum for 24 h. The concentrations of nanoparticles in the polymer matrix were changed as 10, 20 and 30 wt% for ZnS and HBC/ZnS thin films.

The thicknesses and linear absorption spectra of all nanocomposites were measured using spectroscopic ellipsometer and a scanning spectrophotometer. Q-switched Nd:YAG laser (QuantelBrilliant) with 532 nm, 4 ns pulse duration and 10Hz repetition rate was used as a light source for the open aperture (OA) Z-scan measurements. The particle size of each film was specified by the transmission electron microscopy (FEI/G2 S Twin 200 kV RTEM). Inel Equinox 1000 XRD equipped with CoK α radiation was used to determine crystal structure of the ZnS and HBC/ZnS. Rigaku D/Max 2200 ULTIMAN XRD equipped with CuK α radiation was used for the analysis of HBC. EDX pattern of ZnS and HBC/ZnS nanoparticles were obtained by BRUKER AXS XFlash detector 4010 and ZEISS EVO 40 scanning electron microscope.

3. Results and discussions

XRD patterns of ZnS, HBC/ZnS and HBC are shown in Fig. 1. The XRD pattern agree with that of β -ZnS (cubic) reported in the JCPDS powder

diffraction file No.5-0566. ZnS particles show three important diffraction peaks. These peaks are scattering angles of 33.54°, 56.29° and 67.01° agree with the reflections from the (111), (220) and (311) crystal planes of ZnS, respectively [15–17]. HBC/ZnS composite material has same diffraction patterns with ZnS particles. The average grain size is approximately 2.88 nm, which was calculated from the full width at half-maximum (FWHM) of the most intense peak using the Debye–Scherrer's equation [15,16]. XRD patterns of HBC exhibit one broad peak at around 20° indicating the amorphous structure of HBC. XRD patterns of ZnS exhibit more intense peaks comparing HBC due to the high crystalline structure. No diffraction pattern was observed that belongs to HBC in the XRD patterns of HBC/ZnS. This is because of the relatively low content and amorphous structure of HBC [18].

EDX studies were carried out in order to examine the chemical formation and composition of the synthesized nanoparticles and also to confirm the presence of ZnS and carbon-based material (HBC) ZnS. Fig. 2 shows the EDX spectra of ZnS and HBC/ZnS. It was clearly seen from the EDX pattern that the nanoparticles were successfully synthesized.

The linear absorption spectra of ZnS and HBC/ZnS thin films depending on the concentration of nanoparticle in PMMA are given in Figs. 3 and 4, respectively. As seen from Figs. 3 and 4, all nanocomposite films display no absorption in the visible region (400–800 nm). The absorption peaks of nanocomposites are in the wavelength range of 250–340 nm wavelength which corresponds to the band gap energy. The thicknesses of all films are found to be around 1300 nm. The band gap energy values were specified according to the optical absorption spectra. Eq. (1) was used for this purpose,

$$(\alpha h\nu)^{1/n} = A(h\nu - E_g) \quad (1)$$

In this equation, α is the linear absorption coefficient, h is the Planck constant, ν is the frequency and E_g is the energy gap. The value of n for ZnS which has direct band gap is $\frac{1}{2}$ [2,19,20]. The values of E_g are given in Table 1. Observed band gap energy values of the nanocomposite films are higher than that of the bulk value of ZnS (3.68 eV) because of the quantum confinement effect [2,19]. Fig. 5 shows the energy band gap

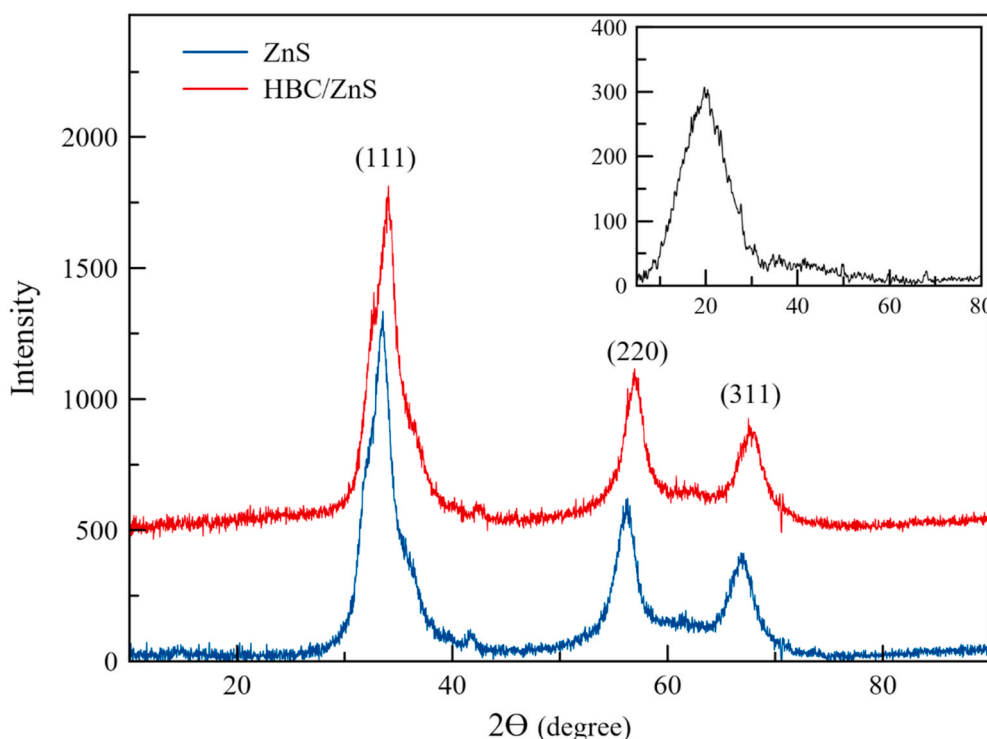


Fig. 1. XRD patterns of ZnS, HBC/ZnS and HBC (inset).

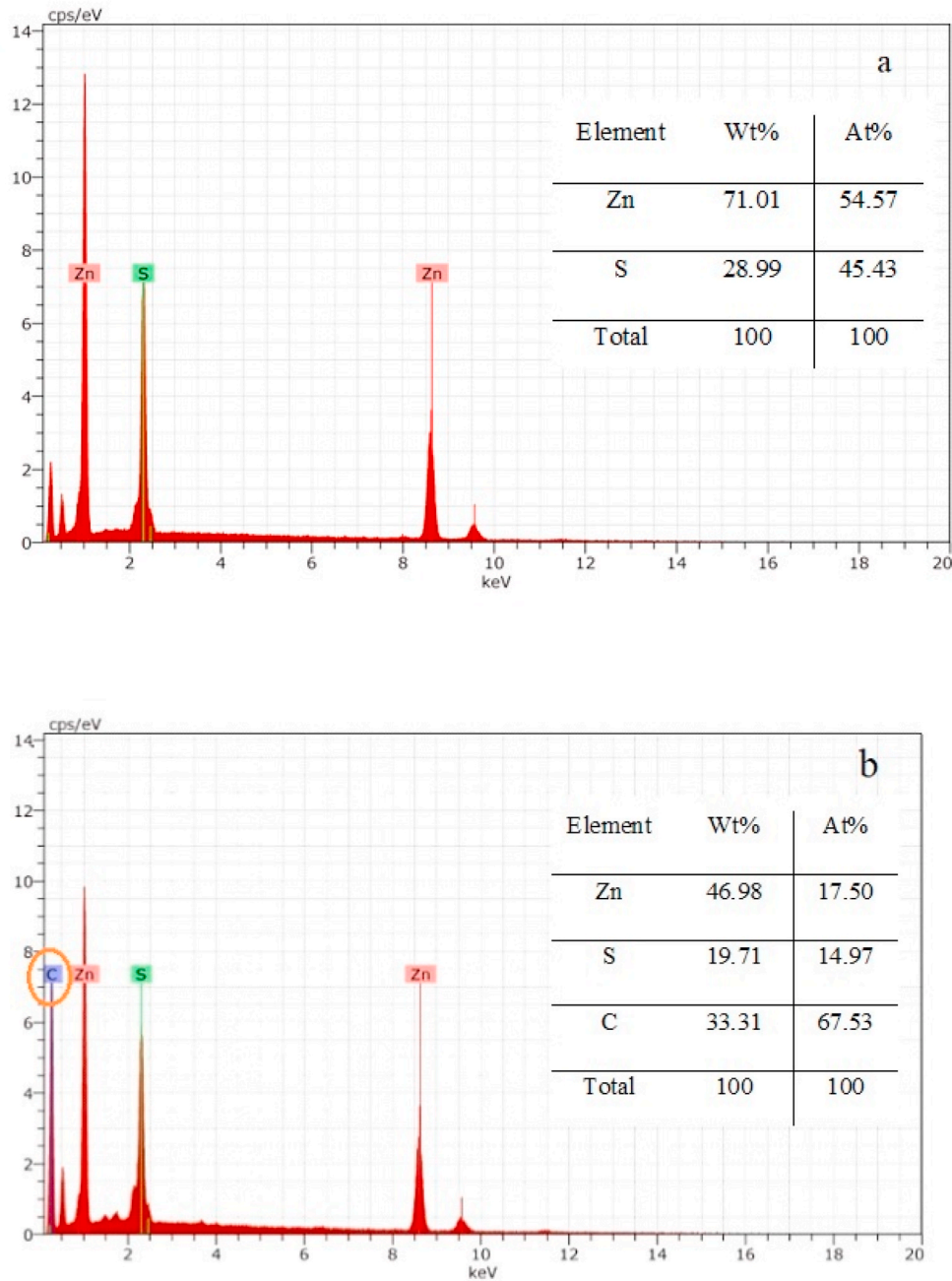


Fig. 2. EDX spectra of (a) ZnS and (b) HBC/ZnS.

spectra with different concentration of nanoparticles. As seen from the figures, the energy band gap value decreases with the increasing of the weight concentration of nanoparticles in PMMA. The relationship between band gap energy and nanoparticle concentration in PMMA is inversely proportional. This situation can be explained by the increasing carrier-carrier interaction leads to the band levels get closer to each other [21,22]. In addition, the band gap energy is decreased due to the enhancement of the density of localized states based on the presence of unsaturated defects [21,23]. Fig. 5 and Table 1 also show that at the same concentration of nanoparticles in polymer, the band gap energy of ZnS nanocomposites is higher than HBC/ZnS. This can be explained by the change of the band gap depending on the nanoparticle size as well as the nanoparticle concentration in the nanocomposite material. The bandwidth decreases as the nanoparticle size increases [24, 25]. Fig. 6 depicts TEM images of ZnS and HBC/ZnS nanoparticles. The addition of carbon caused the ZnS nanoparticles to coalesce. As seen in the figure,

the relatively higher HBC content ZnS nanoparticles were agglomerated in HBC/ZnS composite. ZnS nanoparticles have particle size around 20–25 nm (Fig. 6a).

Figs. 7–10 depict the OA Z scan traces of ZnS and HBC/ZnS nanocomposites at the same input intensity. All of the films exhibited positive nonlinear absorptions and a theoretical model containing one photon (OPA), two photon (TPA) and free carrier absorption (FCA) were used to obtain the nonlinear absorption coefficients and saturation intensity thresholds [26]. The theoretical fits were obtained by using the following Eq. 2

$$\frac{dI}{dz'} = -\frac{\alpha I}{1 + I/I_{SAT}} - \frac{\beta I^2}{1 + I^2/I_{SAT}^2} - \frac{\sigma_0 \Delta N I}{1 + I^2/I_{SAT}^2} \quad (2)$$

In this model, the terms correspond to OPA, TPA, FCA and their saturations, respectively. In Eq. (2), σ_0 is FCA cross section, β is the TPA

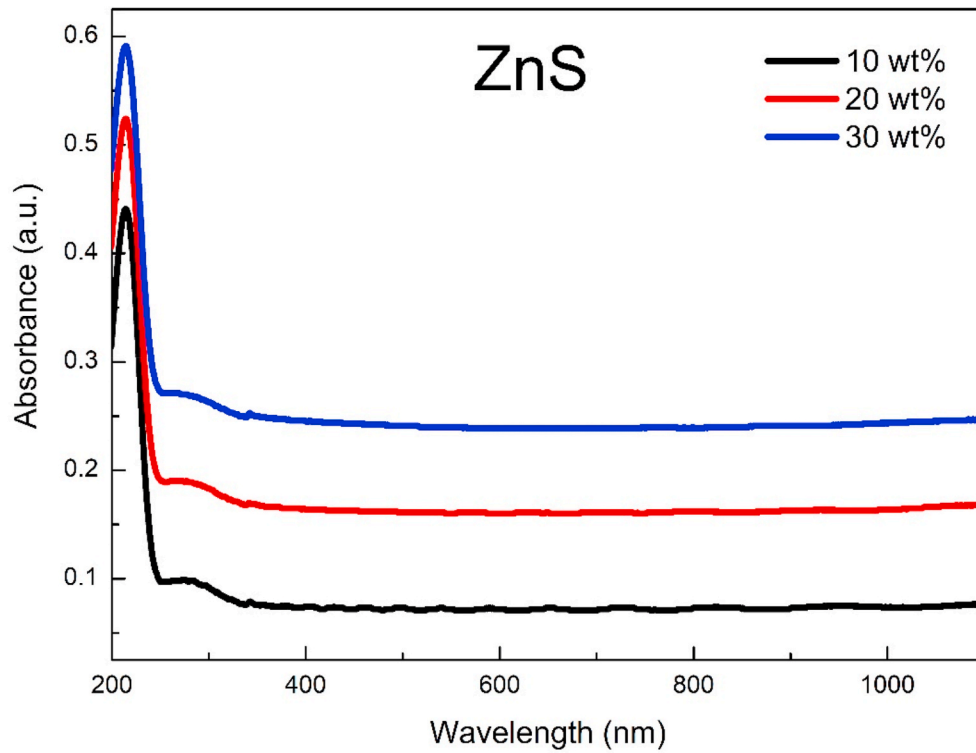


Fig. 3. Linear absorption spectra of ZnS nanocomposite with different concentration of nanoparticles (wt%) in PMMA.

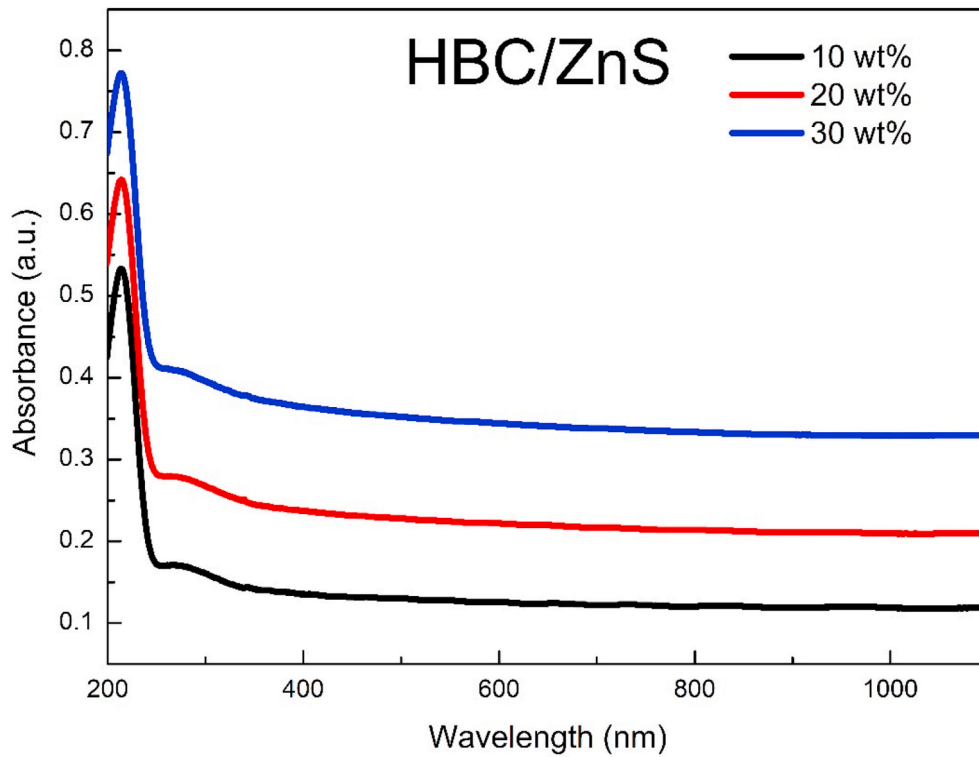


Fig. 4. Linear absorption spectra of HBC/ZnS nanocomposite with different concentration of nanoparticles (wt%) in PMMA.

coefficient, α is the linear absorption coefficient, z' is the propagation distance of light in the material, I_{SAT} is the saturation intensity threshold for SA, ΔN is the photocarrier density and obtained by

$$\Delta N = \left(\sigma \tau_0 / \hbar \omega_0 \right) I \quad (3)$$

Eq. (4) is obtained by substituting Eq. (3) into Eq. 2

Table 1

Band gap energy (E_g) and nonlinear optical parameters of ZnS and HBC/ZnS nanocomposites.

Sample	wt%	E_g (eV)	β_{eff} (cm/GW)	I_{SAT} (GW/cm ²)
ZnS	10	4,71	$9,88 \times 10^3$	3,59
ZnS	20	4,39	$6,92 \times 10^4$	3,73
ZnS	30	3,85	$1,46 \times 10^5$	5,45
HBC/ZnS	10	4,35	$6,64 \times 10^4$	4,42
HBC/ZnS	20	4,14	$9,27 \times 10^4$	5,29
HBC/ZnS	30	3,70	$1,86 \times 10^5$	7,79

$$\frac{dI}{dz'} = -\frac{\alpha I}{1 + I/I_{SAT}} - \frac{\beta_{eff} I^2}{1 + I^2/I_{SAT}^2} = -f(I) \quad (4)$$

Where

$$\beta_{eff} = \beta + (\sigma_0 \alpha \tau_0 / \hbar \omega_0) \quad (5)$$

The Adomian decomposition method provides an approximation for saturable absorption in OA Z-scan [27]. By considering this model, Eq. (4) can be integrated

$$I_{out} = I_{in} - \int_0^L f(I) dz' \quad (6)$$

In this equation, I_{out} is transmitted optical intensity at the exit face of the material, I_{in} is the optical intensity at the input of the material and L is the thickness of the material. The following equation can be used to

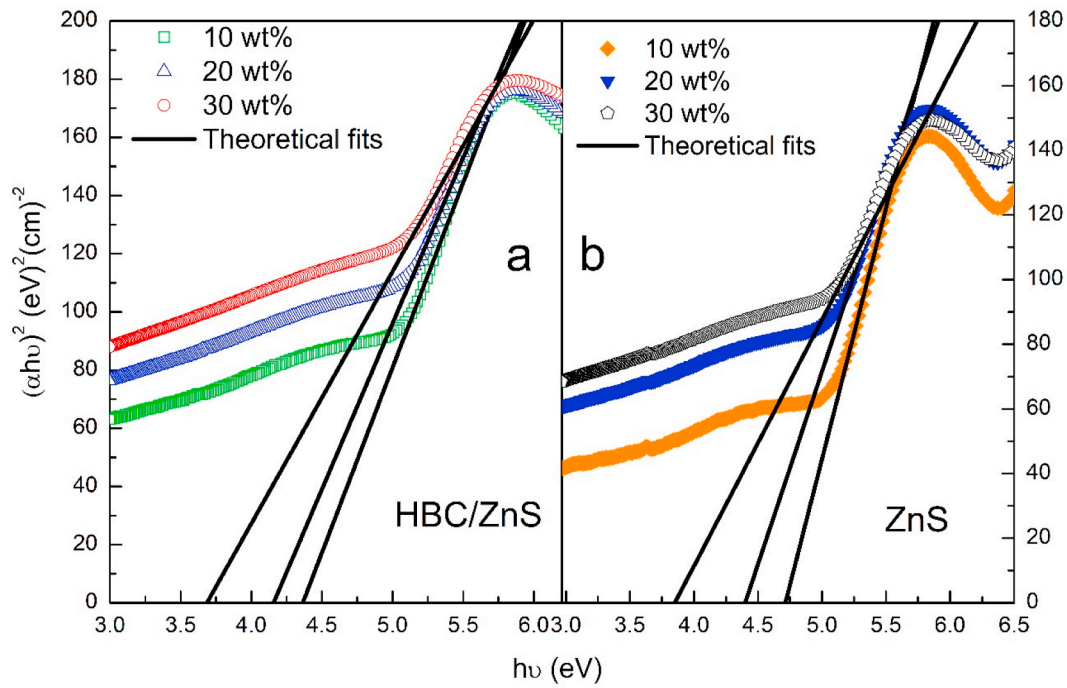


Fig. 5. The energy band gap spectra with 10,20 and 30 wt% in polymer (a)HBC/ZnS(b) ZnS.

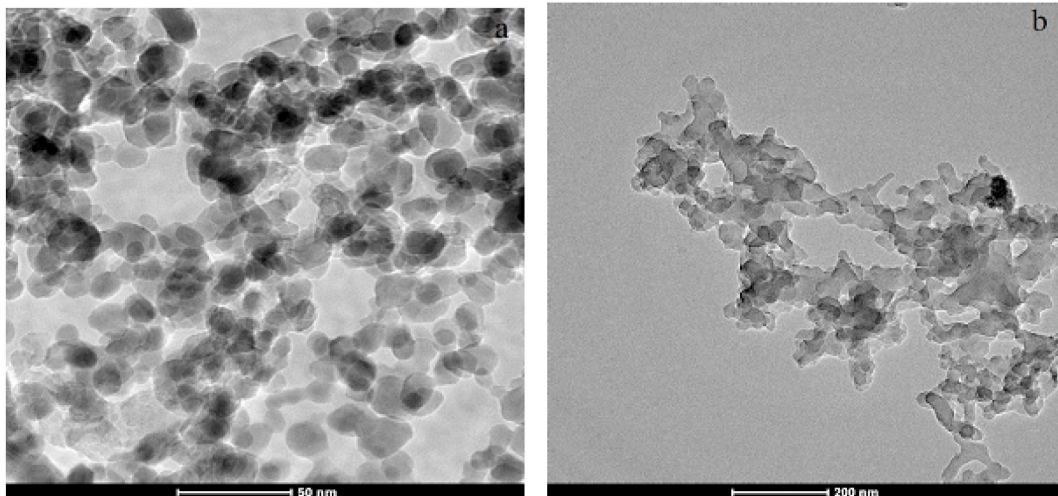


Fig. 6. TEM view(a) ZnS (b) HBC/ZnS.

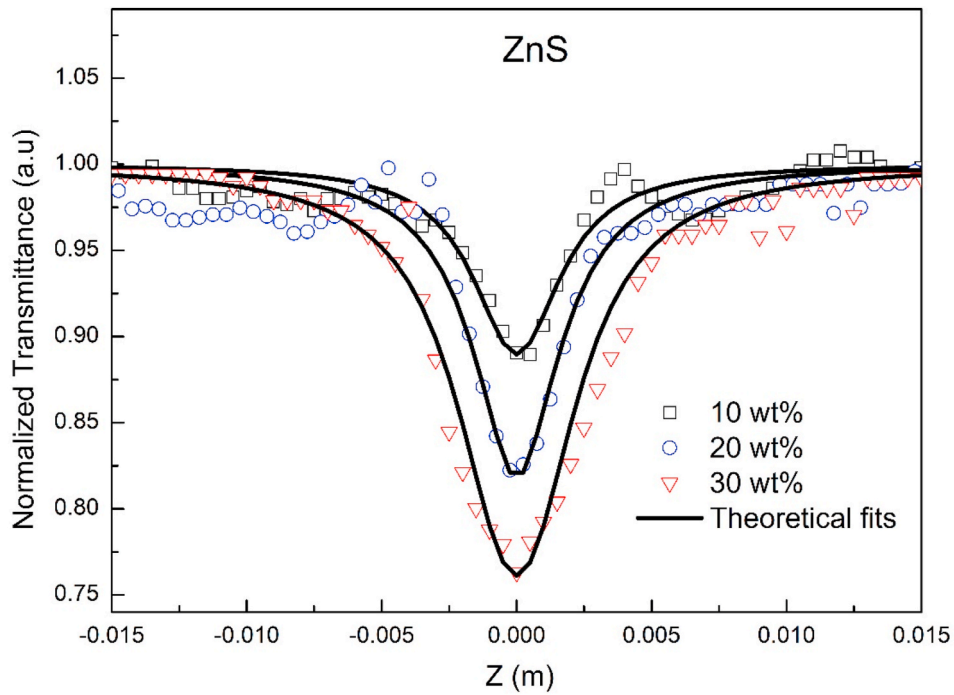


Fig. 7. Z-scan experimental results and theoretical fit for ZnS/PMMA with different concentration of nanoparticles. ($I_0 = 4 \times 10^{11} \text{ W/m}^2$).

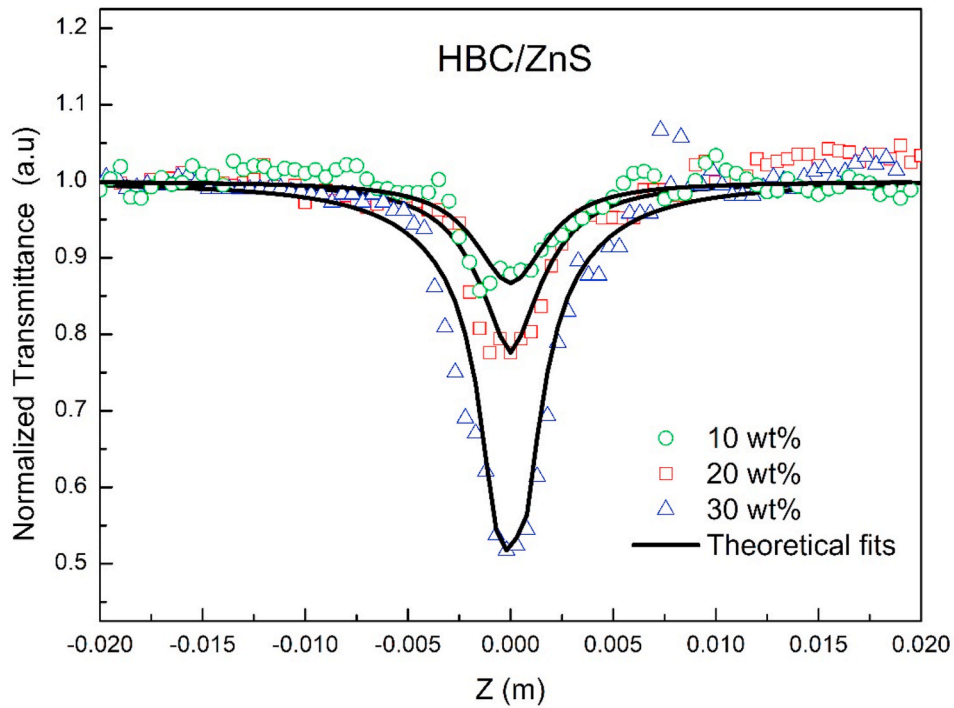


Fig. 8. Normalized transmittance of HBC/ZnS with 10, 20 and 30 wt% in PMMA ($I_0 = 4 \times 10^{11} \text{ W/m}^2$).

obtain the OA Z-scan expression for a pulsed laser.

$$T(x) = \frac{\int_{-\infty}^{\infty} T(x, t) h(t) dt}{\int_{-\infty}^{\infty} h(t) dt} \quad (7)$$

In Eq. (7), $h(t)$ is the temporal profile of the laser pulse and given as $h(t) = \exp[-(t/t_0)^2]$. The fitting results obtained using the above equations are given in Figs. 7–9 and Table 1.

Figs. 7 and 8 show the OA Z-scan results as a function of the

nanoparticle concentration in polymer (wt%) for both ZnS and HBC/ZnS thin films. As seen from Table 1 and Figs. 7 and 8, the high concentration of nanoparticles exhibits stronger nonlinear absorption than low concentration. Based on the results of the band energy given in Fig. 5, the reduction of the forbidden energy with the nanoparticle concentration enhances non-linear absorption [28,29]. Fig. 9 depicts the nonlinear absorption coefficient (β_{eff}) at the different nanoparticle concentration in PMMA (wt %). It is clearly seen that the β_{eff} value increases with addition of nanoparticles. Similar behavior was also reported in the

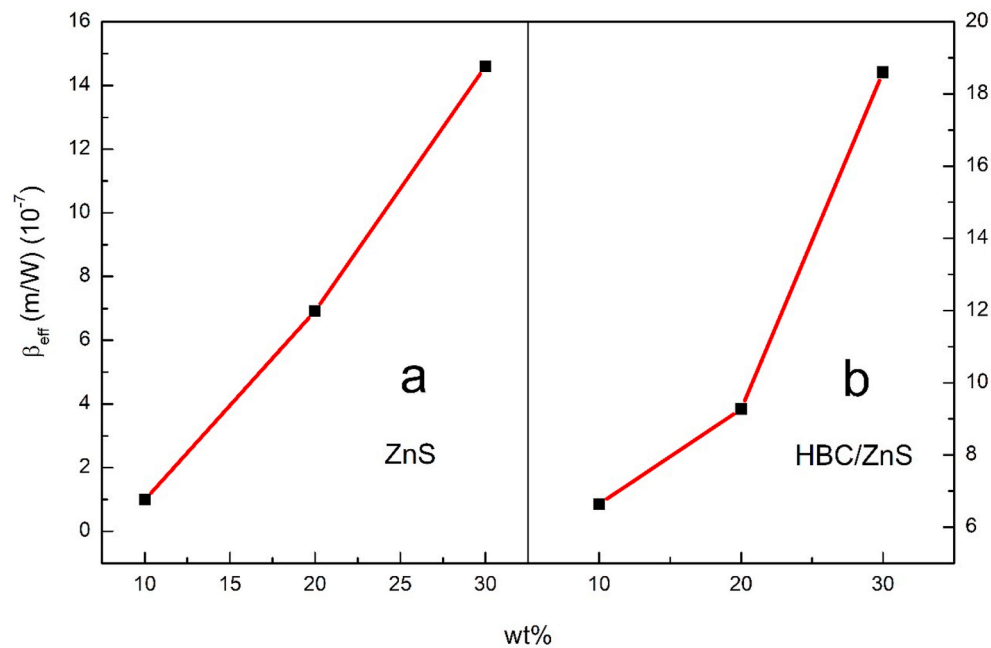


Fig. 9. The nonlinear absorption coefficient (β_{eff}) versus on-axis nanoparticle concentration in PMMA (wt %) (a) ZnS (b) HBC/ZnS.

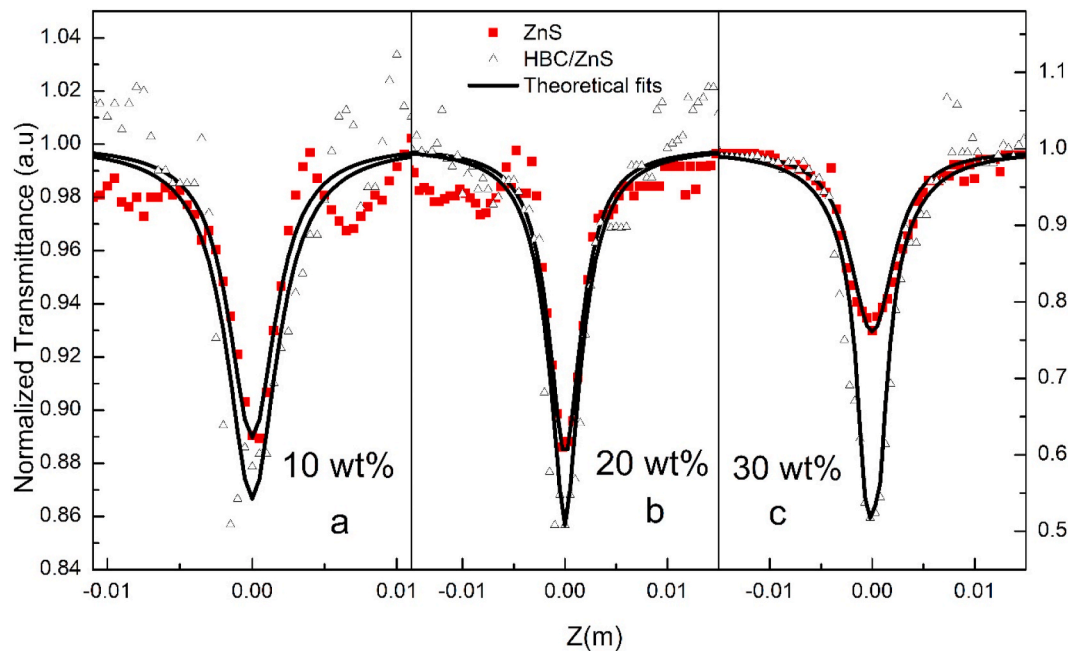


Fig. 10. Normalized transmittance of ZnS and HBC/ZnS nanocomposites at the same nanoparticle concentration in PMMA (a) 10 wt% (b) 20 wt% and (c) 30 wt% ($I_0 = 4 \times 10^{11}$ W/m²).

literature [28–32].

As seen in Fig. 10, the HBC/ZnS films shows an enhancement in the value of two photon absorption than that of ZnS for the same concentration of nanoparticles (wt%). There is an increase in β_{eff} value due to the nanoparticles aggregates (Fig. 6b) [33].

In our previous studies, the nonlinear absorption and saturable absorption properties of amorphous semiconductor films depend on the thickness and doping were investigated [26,34–38]. Since these films have defects such as dangling bonds and impurities, more localized defect states are observed in the band gap. Our results showed that the increasing defect states due to doping and thickness caused the

nonlinear and saturable absorption behaviors to change. The saturable absorption was observed instead of the nonlinear absorption as the thickness decreased at 1064 nm (ns and ps pulse duration) and 800 nm (fs pulse duration). The β_{eff} and I_{SAT} values of these semiconductor films are in the order of 10^3 – 10^6 cm/GW and 10^{-3} – 10^{-1} GW/cm², respectively. The nonlinear absorption coefficients are sensitive to the defect states and can be controlled by changing the doping, concentration and film thicknesses. In addition to that, we also studied the nonlinear optical properties of the undoped and doped TiO₂, SnO, CdO and ZnO nanostructure films [9,39–41]. The β_{eff} and I_{SAT} values for these undoped nanostructure films is in the order of 10^5 cm/GW and around 5

GW/cm², respectively. It is interesting to compare β_{eff} and I_{SAT} values of the present study with the values of our previous thin film studies. The β_{eff} value of the ZnS (10 wt%) and HBC/ZnS (10 wt%) nanocomposite films are found to be 9.88×10^3 cm/GW and 6.64×10^4 cm/GW, respectively. It is increased up to 1.46×10^5 cm/GW and 1.86×10^5 cm/GW with the variation of nanoparticle concentration in PMMA and the addition of carbon. These β_{eff} values of the studied films are slightly higher than our previous results of SnO, CdO and ZnO nanostructure films [39–41]. This situation can be attributed to the decreasing energy band gap values and the aggregation of the nanoparticles in PMMA.

4. Conclusion

ZnS and HBC/ZnS nanocomposites with different concentrations of nanoparticles in PMMA were prepared. The nanoparticle concentration effect on linear and nonlinear optical absorption properties was investigated with various concentrations (10, 20 and 30 wt %). The obtained band gap energies of both ZnS and HBC/ZnS films are higher than that of the bulk value. As a result of the quantum confinement effect, the blue shift is observed in our bandgap results. The addition of more nanoparticles to nanocomposites decreased the band gap energy of all studied samples. It has been observed that nonlinear optical properties can be improved by increasing the weight percent of nanoparticles in the nanocomposite material prepared with PMMA polymer. Our experimental results also showed that the band gap energy and nonlinear absorption coefficient depend on the nanoparticles aggregates in the nanocomposites.

CRediT authorship contribution statement

Sezen Tekin: Investigation, Writing - review & editing, Data curation. **Ahmet Karatay:** Supervision, Conceptualization, Methodology, Writing - review & editing. **Yusuf Osman Donar:** Resources, Investigation. **Selva Bilge:** Resources, Investigation. **Elif Akhuseyin Yildiz:** Investigation, Data curation. **Ali Sinag:** Conceptualization, Methodology. **Ayhan Elmali:** Supervision, Conceptualization, Methodology, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] Z. Dehghani, S. Nazerdeylami, E. Saievar-Iranizad, M.H.M. Ara, J. Phys. Chem. Solid. 72 (2011) 1008–1010.
- [2] C.S. Pathak, D.D. Mishra, V. Agarwala, M.K. Mandal, Ceram. Int. 38 (2012) 6191–6195.
- [3] Y.Z. Li, S.Y. Zhang, L.Z. Gao, W.J. Chen, L.J. Gao, W.K. Zhang, J.L. Cui, S.H. Yan, Synth React Inorg M 44 (2014) 942–945.
- [4] J. Osuntokun, P.A. Ajibade, J. Nanomater. (2016) 2016.
- [5] R.S. Meshram, Thombre, R. M., Structural and optical properties of zns thin films deposited by spray pyrolysis technique. IJCA Proceedings on International Conference on Benchmarks in Engineering Science and Technology 2012 ICBEST (3):22–25, October 2012.
- [6] K. Ozga, J. Michel, B.D. Nechyporuk, J. Ebothe, I.V. Kityk, A.A. Albassam, A.M. El-Naggar, A.O. Fedorchuk, Physica E 81 (2016) 281–289.
- [7] M.C. Divyasree, K. Chandrasekharan, Opt. Mater. 67 (2017) 119–124.
- [8] K.P. Tiwary, S.K. Choubey, K. Sharma, Chalcogenide Lett. 10 (2013) 319–323.
- [9] A. Karatay, Y.O. Donar, A. Sinag, A. Elmali, Optic Laser. Technol. 108 (2018) 510–514.
- [10] X. Qi, N. Liu, Y. Lian, RSC Adv. 5 (2015) 17526–17531.
- [11] Y.O. Donar, E. Caglar, A. Sinag, Fuel 183 (2016) 366–372.
- [12] D. Mohan, A. Sarswat, Y.S. Ok Jr., CU. Pittman, Bioresour Technol 160 (2014) 191–202.
- [13] E. Lam, J.H.T. Luong, ACS Catal. 4 (2014) 3393–3410.
- [14] Y.O. Donar, S. Bilge, A. Sinag, O. Pliekhov, ChemCatChem 10 (2018) 1134–1139.
- [15] K. Vijai Anand, M. Karl Chinnu, R. Mohan Kumar, R. Mohan, R. Jayavel, Appl. Surf. Sci. 255 (2009) 8879–8882.
- [16] K. Vijai Anand, M. Karl Chinnu, R. Mohan Kumar, R. Mohan, R. Jayavel, J. Alloys Compd. 496 (2010) 665–668.
- [17] K. Vijai Anand, R. Mohan, R. Mohan Kumar, M. Karl Chinnu, R. Jayavel, Int. J. Nanosci. 10 (2011) 487–493.
- [18] Y.O. Donar, S. Bilge, A. Sinag, Mater. Sci. Semicond. Process. 118 (2020).
- [19] J.P. Borah, J. Barman, K.C. Sarma, Chalcogenide Lett. 5 (2008) 201–208.
- [20] P.E. Agbo, P.A. Nwofe, L.O. Odo, Chalcogenide Lett. 14 (2017) 357–363.
- [21] H.M. Shanshool, M. Yahaya, W.M.M. Yunus, I.Y. Abdullah, J. Mater. Sci. Mater. Electron. 27 (2016) 9804–9811.
- [22] L.M. Iriripan, Spectral and Nonlinear Optical Characterization of ZnO Nanocomposites, PhD Thesis, Cochin University of Science and Technology, 2008.
- [23] M.A.R.H. El-Zahed, A. El-Korashy, Vacuum 68 (2003) 19–27.
- [24] A.H. Yuwono, J.M. Xue, J. Wang, H.I. Elim, W. Ji, Y. Li, T.J. White, J. Mater. Chem. 13 (2003) 1475–1479.
- [25] C. Sciancalepore, T. Cassano, M.L. Curri, D. Mecerreyes, A. Valentini, A. Agostiano, R. Tommasi, M. Striccoli, Nanotechnology 19 (2008).
- [26] U. Kurum, M. Yuksek, H.G. Yaglioglu, A. Elmali, A. Ates, M. Karabulut, G. Mamedov, J. Appl. Phys. 108 (2010).
- [27] B. Gu, Y.X. Fan, J. Wang, J. Chen, J.P. Ding, H.T. Wang, B. Guo, Phys Rev A 73 (2006).
- [28] H.M. Shanshool, M. Yahaya, W.M.M. Yunus, I.Y. Abdullah, J. Teknol 78 (2016) 33–38.
- [29] R.L. Thomas, Vasuja, M. Hari, V.P.N. Nampoori, P. Radhakrishnan, S. Thomas, J. Optoelectron. Adv. Mater. 13 (2011) 523–527.
- [30] P.C. Haripadman, M.K. Kavitha, H. John, B. Krishnan, P. Gopinath, Appl. Phys. Lett. 101 (2012).
- [31] R. Sreeja, J. John, P.M. Aneesh, M.K. Jayaraj, Optic Commun. 283 (2010) 2908–2913.
- [32] L. Iriripan, V.P.N. Nampoori, P. Radhakrishnan, B. Krishnan, A. Deepthy, J. Appl. Phys. 103 (2008).
- [33] Y. Liu, X.L. Wang, M.Q. Wei, H. Wang, Y.P. Tian, S.K. Li, Z.M. Xue, J.X. Yang, L. Kong, Opt. Mater. 62 (2016) 485–493.
- [34] M. Yuksek, U. Kurum, H.G. Yaglioglu, A. Elmali, A. Ates, J. Appl. Phys. 107 (2010).
- [35] A. Karatay, H.G. Yaglioglu, A. Elmali, M. Parlak, H. Karaagac, Optic Commun. 285 (2012) 1471–1475.
- [36] A. Karatay, B. Kucukoz, G. Cankaya, A. Ates, A. Elmali, Opt. Mater. 73 (2017) 20–24.
- [37] A. Karatay, C. Aksoy, H.G. Yaglioglu, A. Elmali, U. Kurum, A. Ates, N. Gasanly, J Optics-Uk 13 (2011).
- [38] P. Isik, A. Karatay, H.G. Yaglioglu, A. Elmali, U. Kurum, A. Ates, N. Gasanly, Optic Commun. 288 (2013) 107–113.
- [39] Y. Pepe, M.A. Yildirim, A. Karatay, A. Ates, H. Unver, A. Elmali, Opt. Mater. 98 (2019).
- [40] Y. Pepe, A. Karatay, Y.O. Donar, A. Sinag, H. Unver, A. Elmali, Opt. Mater. 103 (2020).
- [41] Y. Pepe, A. Karatay, Y.O. Donar, S. Bilge, A. Elmali, A. Sinag, Opt. Mater. 101 (2020).